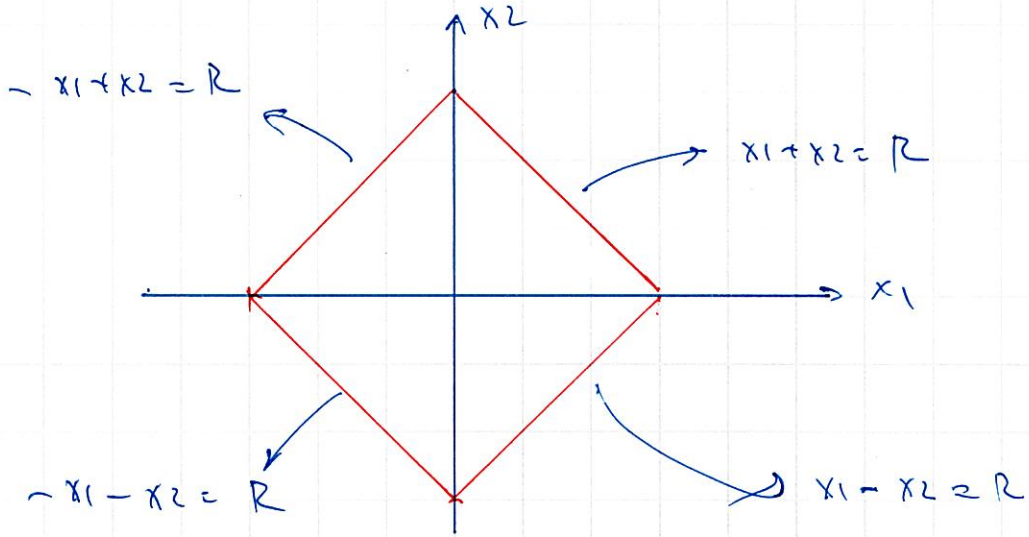
Projections onto the L1-Ball

- Consider the conc $X = \{ x \in \mathbb{R}^n : \sum_{i=1}^{n} |x_i| \leq R \}$



- We want to find $\Pi_X(\hat{y})$, where $\hat{y}$ is a fixed point

- We want to solve the problem

$$\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$$

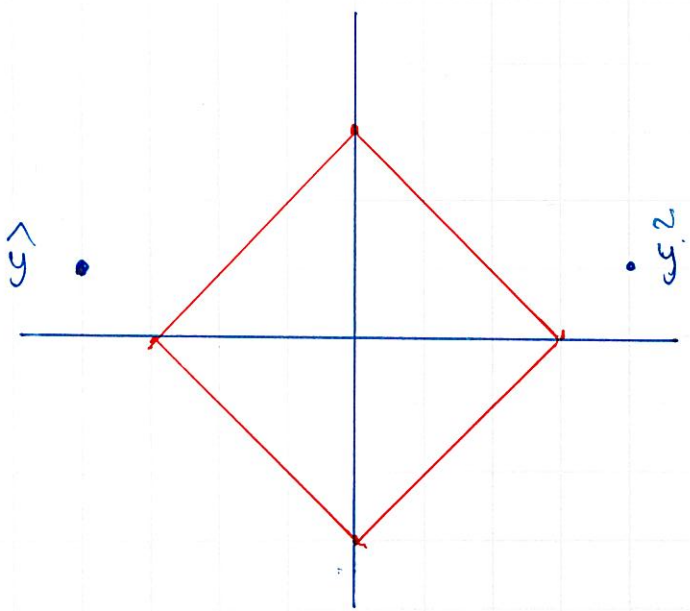
st

$$\sum_{i=1}^{n} |x_i| \leq R$$

Assume that $\hat{y} \notin X$

Observation 1:

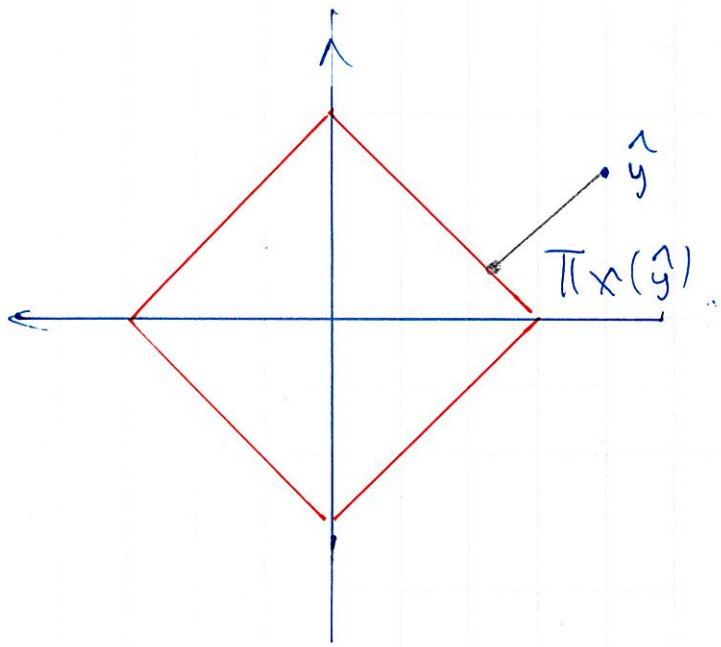
We can assume $\hat{y}_i \geq 0 \quad i=1,\dots,n$



Projecting $\hat{y}$ onto the $L1$-ball is same as projecting $\tilde{y}$ and negating the x-coordinate of the projected point

observation 2:

given $\hat{y}$ such that $\hat{y}_i \geq 0$ $i=1,\dots,n$, the projection of $\hat{y}$ onto the $L1$-ball has all non-negative components



Let's mathematically establish that both of these observations are indeed correct

Observation 1:

- When projecting $\hat{y}$ onto L1-Ball, we can assume $\hat{y}_i \geq 0$ $\forall i = 1, \dots, n$.
Why?

Assume $\hat{y}_j < 0$:

$$\min \sum_{i \neq j} (x_i - \hat{y}_i)^2 + (x_j - \hat{y}_j)^2 \quad (*)$$

st

$$\sum_{i \neq j} |x_i| + |x_j| \leq R$$

Consider instead this problem:

$$\min \sum_{i \neq j} (x_i - \hat{y}_i)^2 + (x_j - (-\hat{y}_j))^2 \quad (\dagger)$$

st

$$\sum_{i \neq j} |x_i| + |x_j| \leq R$$

Let's argue both problems have the same opt. obj. value.

- Let x^* be an opt. solution to problem $(*)$. Define $\tilde{x}$ such that
$\tilde{x}_i = x^*_i$ $i \neq j$, $\tilde{x}_j = -x^*_j$

Because x^* is opt. to $(*)$, we have $\sum_{i=1}^{n} |x_i^*| \leq R$ but also $|x_i^*| = |\tilde{x}_i|$ for all $i = 1, \dots, n$.
Thus $\tilde{x}$ is feasible to $(\dagger)$.

The obj. value that $\tilde{x}$ provides for problem $(\ddagger)$ is

$$\sum_{i \neq j} (\tilde{x}_i - \hat{y}_i)^2 + (\tilde{x}_j - (-\hat{y}_j))^2 = \sum_{i \neq j} (x_i^* - \hat{y}_i) + (-x_j^* + \hat{y}_j)^2$$

$$= \sum_{i \neq j} (x_i^* - \hat{y}_i) + (x_j^* - \hat{y}_j)^2,$$

which is the opt. objective value of problem $(*)$

- Thus, given an opt. solution to $(*)$, we can construct a feasible solution to $(\ddagger)$ with obj. value equal to the opt. obj. value of $(*)$

This means that the opt. obj. value of $(\ddagger)$ is less than or equal to the opt obj. value of $(*)$.

- We can follow precisely the reverse argument to show that given a optimal solution to $(\ddagger)$, we can construct a feasible solution to $(*)$ with an obj. value equal to the opt. obj. value of $(\ddagger)$

This means that the opt. obj value of $(*)$ is less than or equal to the opt. obj. value of $(\ddagger)$

- Thus, the two problems have the same opt. obj. values. Done!

Observation 2:

Let $\hat{y}$ be such that $\hat{y}_i \geq 0$ $i=1,\dots,n$. Then the projection of $\hat{y}$ onto the $L1$-ball all has non-negative components. Why?

- To get a contradiction, let $\hat{y}$ be such that $\hat{y}_i \geq 0$ $\forall i=1,\dots,n$, but the j-th component of the projection of $\hat{y}$ onto $L1$-ball is negative.

$$\min \quad \sum_{i \neq j} (x_i - \hat{y}_i)^2 + (x_j - \hat{y}_j)^2 \qquad (**)$$

$$\text{st} \quad \sum_{i \neq j} |x_i| + |x_j| \leq R$$

- Let x^* be an opt. solution to $(**)$ and assume $x_j^* < 0$.
- Define the solution $\tilde{x}$ to $(**)$ as $\tilde{x}_i = x_i^*$ for $i \neq j$, $\tilde{x}_j = 0$
- Because x^* is feasible to $(**)$, we have $\sum_{i \neq j} |x_i^*| + |x_j^*| \leq R$, but then

$$\sum_{i \neq j} |\tilde{x}_i| + |\tilde{x}_j| = \sum_{i \neq j} |x_i^*| + 0 \leq R.$$

So $\tilde{x}$ is feasible to $(**)$

- The solution $\tilde{x}$ provides the following objective value for problem (**)

$$\sum_{i \neq j} (\tilde{x}_i - \hat{y}_i)^2 + (\tilde{x}_j - \hat{y}_j)^2 = \sum_{i \neq j} (x_i^* - \hat{y}_i)^2 + (\hat{y}_j)^2 < \sum_{i \neq j} (x_i^* - \hat{y}_i)^2 + (\hat{y}_j - x_j^*)^2$$

$$= \sum_{i \neq j} (x_i^* - \hat{y}_i) + (x_j^* - \hat{y}_j)^2$$

Thus, $\tilde{x}$ gives a strictly better obj. value for problem (**) than the solution x^*.

- So the solution x^* cannot be an opt. solution to problem (**)

Contradiction! done!

- We want to solve the problem

$$\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$$
$$\text{st} \quad \sum_{i=1}^{n} |x_i| \leq R$$

- By observation 1, we can equivalently solve

$$\min \sum_{i=1}^{n} (x_i - |\hat{y}_i|)^2$$
$$\text{st} \quad \sum_{i=1}^{n} |x_i| \leq R$$

- By observation 2, we can further equivalently solve

$$\min \sum_{i=1}^{n} (x_i - |\hat{y}_i|)^2$$
$$\text{st} \quad \sum_{i=1}^{n} x_i \leq R$$
$$x_i \geq 0 \quad \forall i = 1, \dots, n$$

- when $\hat{y} \notin X$, (which is the interesting case), the last problem is equivalent to

$$\min \sum_{i=1}^{n} (x_i - |\hat{y}_i|)^2$$

$$\text{st} \quad \sum_{i=1}^{n} x_i = R$$

$$x_i \geq 0 \quad \forall i = 1, \dots, n$$

- By the detour to last lecture, to solve the problem above, find λ^* such that

$$\sum_{i=1}^{n} \left[|\hat{y}_i| - \frac{\lambda^*}{2} \right]^+ = R$$

then the opt. solution to the problem above is

$$x_i^* = \left[|\hat{y}_i| - \frac{\lambda^*}{2} \right]^+$$

Detour to the last lecture: Projecting onto the probability simplex.

$$\min \quad \sum_{i=1}^{n} (x_i - \hat{y}_i)^2 \qquad (***)$$

st

$$\sum_{i=1}^{n} x_i = \cancel{1}\ R$$

$$x_i \geq 0 \quad \forall i = 1, \dots, n$$

To solve the problem above, find λ^* such that

$$\sum_{i=1}^{n} \left[\hat{y}_i - \frac{\lambda^*}{2} \right]^+ = \cancel{1}\ R$$

Then the opt solution to problem $(***)$ is given by

$$x_i^* = \left[\hat{y}_i - \frac{\lambda^*}{2} \right]^+$$

- To conclude, let $\hat{y}$ have components with arbitrary signs and $\hat{y} \notin X$. We want to solve

$$\min \sum_{i=1}^{n} (x_i - \hat{y}_i)^2$$
$$\text{st} \quad \sum_{i=1}^{n} |x_i| \leq R$$

- Find λ^* such that $\displaystyle\sum_{i=1}^{n} \left[|\hat{y}_i| - \frac{\lambda^*}{2} \right]^+ = R$

- Thus, the opt. solution x^* to the lost problem above is given by

$$x_i^* = \begin{cases} \left[|\hat{y}_i| - \frac{\lambda^*}{2} \right]^+ & \hat{y}_i \geq 0 \\ -\left[|\hat{y}_i| - \frac{\lambda^*}{2} \right]^+ & \hat{y}_i < 0 \end{cases}$$